Contents

1 Combinatorics **2**

2 Conditional probability, independence and Bayes' rule **5**

3 Random variables and discrete distributions **9**

4 Dependence, sums and generating functions **13**

English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Notes

Each topic contains *core problems*, covering the minimum required for that topic; *additional problems*, providing variety and extending the core material; and *challenges*, related to the topic but going beyond the requirements of standard introductory courses in probability and statistics. The following symbols are used:

- a problem introducing fundamental objects of the course;
- a challenging problem;
- an interesting example;
- a problem from an FMI test;
- an independent research task. All resources are permitted, including literature, software and AI. You are expected to check the results, understand them, and be able to explain them.

Note on the use of AI. The latest publicly available versions of ChatGPT and Claude (at the time of writing, sol 6.1 and opus 5.5, respectively) were used for formatting, editing and clarifying individual formulations. The compiler is responsible for the final version and its mathematical correctness.

1. Combinatorics

Core problems

Problem 1.1. For $n, k \geq 1$, recall:

(a) how many solutions does $x_1 + \cdots + x_n = k$ have when $x_1, \ldots, x_n$ are positive integers? What if they are nonnegative integers?

(b) state the inclusion–exclusion principle.

Problem 1.2. Let M be a set with n elements. In how many ways can we choose:

(a) k distinct elements of M;

(b) an ordered k-tuple of distinct elements of M;

(c) an ordered k-tuple of elements of M, with repetitions allowed?

Problem 1.3. In how many ways can all k distinguishable particles be distributed among n distinct cells if:

(a) each cell contains at most one particle;

(b) cells may contain any number of particles;

(c) no cell is empty?

Answer the same questions for indistinguishable particles. Also state when the answer is zero.

Problem 1.4. Five distinguishable balls are distributed among three distinct boxes A, B, C. Find the number of distributions in which:

(a) A is empty;

(b) only A is empty;

(c) exactly one box is empty;

(d) at least one box is empty;

(e) no box is empty.

Problem 1.5. Consider the following questions.

(a) In how many ways can we go from $(0, 0)$ to (n, n) on the square lattice using unit steps to the right and upwards? What if the path never goes above the diagonal?

(b) How many correctly balanced parenthesis expressions have n opening and n closing parentheses? Construct an explicit bijection between them and the paths in the second part of the preceding question.

(c) Find simple asymptotic equivalents for the total number of paths and for the number that never go above the diagonal, as $n \to \infty$.

Additional problems

Problem 1.6. How many four-digit numbers can be formed from the digits $1, 2, 3, 4, 5$ if:

(a) repetition is not allowed;

(b) repetition is allowed;

(c) repetition is not allowed and the number is odd?

Problem 1.7. In how many ways can a four-person delegation be selected from 12 candidates, including distinct candidates A, B, C, D, if:

(a) there are no restrictions;

(b) A and B must not both participate;

(c) either both C and D participate, or neither does?

Problem 1.8. Prove combinatorially, and then algebraically, that

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n \qquad \text{and} \qquad \binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

Problem 1.9. Let $p(n)$ be the number of partitions of n into distinct positive integers, and $q(n)$ the number of partitions into odd positive integers, where repetitions are allowed and the order of the summands does not matter. For example, for $n = 6$:

$p(6) = 4:\ 6,\ 5+1,\ 4+2,\ 3+2+1;$ $\qquad q(6) = 4:\ 5+1,\ 3+3,\ 3+1+1+1,\ 1+1+1+1+1+1.$

n	1	2	3	4	5	6	7	8
$p(n)$	1	1	2	2	3	4	5	6
$q(n)$	1	1	2	2	3	4	5	6

By comparing generating functions, prove that $p(n) = q(n)$ for every n.

Challenges and independent research

★ **Problem 1.10.** *(Derangements)* Let D_n be the number of permutations $\sigma \in S_n$ such that $\sigma(i) \neq i$ for every i.

(a) Prove that

$$D_n = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

(b) Prove combinatorially that, for $n \geq 2$,

$$D_n = (n-1)(D_{n-1} + D_{n-2}),$$

where $D_0 = 1$ and $D_1 = 0$.

(c) Prove that, for $n \geq 1$, D_n is the nearest integer to $n!/e$, without using a numerical approximation to e.

★ **Problem 1.11.** *(Necklaces and Burnside's lemma)* Let $X = \{0,1\}^n$ be the set of binary words of length n, and let $G = \{r^0, r^1, \ldots, r^{n-1}\} \cong \mathbb{Z}_n$ be the group of cyclic rotations, $r(x_1 x_2 \ldots x_n) = x_n x_1 \ldots x_{n-1}$. A *necklace* is an orbit of G in X: two words are identified if one is obtained from the other by a cyclic rotation. For example, 0011, 0110, 1100, 1001 form one necklace, and 0101, 1010 form another. Rotations preserve the number of ones, so G also acts on

$$X_k = \{x \in X :\ x \text{ has exactly } k \text{ ones}\}.$$

Write $N(n,k)$ for the number of orbits in X_k, that is, the number of necklaces with n beads and exactly k ones. For $g \in G$, let

$$\mathrm{Fix}(g) = \{x \in X_k :\ g(x) = x\},$$

and let X_k/G be the set of orbits. Use Burnside's (Cauchy–Frobenius) lemma

$$\#(X_k/G) = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

Explain why a rotation g whose cycles on the positions $\{1, \dots, n\}$ have length d can fix a word with k ones only if $d \mid k$. Prove that in this case

$$|\operatorname{Fix}(g)| = \binom{n/d}{k/d},$$

whereas if $d \nmid k$, then $\operatorname{Fix}(g) = \emptyset$. Also prove that for every $d \mid n$ there are exactly $\varphi(d)$ rotations whose cycles have length d. Deduce that, for $1 \le k \le n-1$,

$$N(n,k) = \frac{1}{n} \sum_{d \mid \gcd(n,k)} \varphi(d) \binom{n/d}{k/d},$$

where φ is Euler's totient function. As a check, find $N(12, 4)$.

Problem 1.12. *(Random mappings and cyclic vertices)* Let $f : [n] \to [n]$ be chosen uniformly from all n^n mappings. Let G_f be the directed graph with edges $i \to f(i)$, and K_n the number of vertices lying on a cycle in G_f. You may use AI to propose solutions, acting as their *reviewer*: verification, rather than generation, is assessed.

(a) Use software to find the distribution $\mathbb{P}(K_n = k)$ for $n \le 7$. The code may be AI-generated, but run it and check that $\sum_k \#\{K_n = k\} = n^n$; also check the case $n = 3$ by hand.

(b) Find and prove a formula for $\#\{f : K_n = k\}$. You may ask AI for a proposal. (i) Compare it with the table from part 1. (ii) Write the proof in your own words, separating the choice of cyclic vertices, the permutation on them, and the directed forest rooted at them. (iii) Give your own proof of the formula $k\, n^{n-k-1}$ for the number of forests with k prescribed roots. If you use AI, check whether it has actually proved the formula, and give a proof by a different method. (iv) Find and numerically check the number of mappings with $K_n = k$ and exactly c cycles.

(c) Derive $\mathbb{E}K_n$ in two ways: from the distribution and by linearity, computing $\mathbb{P}(\text{vertex 1 is cyclic})$. Prove the identity between the two expressions. Use the exact formula to calculate $\mathbb{E}K_n$ for $n = 10, 10^2, 10^3, 10^4, 10^5$ and investigate numerically whether

$$\frac{\mathbb{E}K_n}{\sqrt{n}} \longrightarrow \sqrt{\frac{\pi}{2}}.$$

Explain the origin of the constant $\sqrt{\pi/2}$. Optionally investigate the finer asymptotic claim

$$\mathbb{E}K_n - \sqrt{\frac{\pi n}{2}} \longrightarrow -\frac{1}{3}.$$

Let T_n be the number of distinct vertices in the trajectory $1, f(1), f^2(1), \dots$ before its first repetition. Do K_n and T_n have the same distribution, or only the same expectation?

Lean. Let $A, B \subseteq \Omega$, with indicators taking values in $\mathbb{N}$. Define the indicator by

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A, \end{cases}$$

and prove in Lean 4 with Mathlib that, for every $x \in \Omega$,

$$\mathbf{1}_{A \cup B}(x) + \mathbf{1}_{A \cap B}(x) = \mathbf{1}_A(x) + \mathbf{1}_B(x).$$

2. Conditional probability, independence and Bayes' rule

Core problems

Problem 2.1. Two standard dice are rolled independently.

(a) What is the probability that the sum is less than 8, given that it is odd? Are the events "the sum is less than 8" and "the sum is odd" independent?

(b) Are "the sum is 7" and "the first die shows 3" independent? What about "the sum is 8" and "the first die shows 3"?

Problem 2.2. Let A and B be events with $\mathbb{P}(A) > 0$ and $0 < \mathbb{P}(B) < 1$.

(a) Prove that if A and B are independent, then so are A and $\overline{B}$, and also $\overline{A}$ and $\overline{B}$.

(b) Prove that the following four conditions are equivalent:

$$\mathbb{P}(A \cap B) > \mathbb{P}(A)\mathbb{P}(B), \quad \mathbb{P}(A \mid B) > \mathbb{P}(A), \quad \mathbb{P}(B \mid A) > \mathbb{P}(B), \quad \mathbb{P}(A \mid B) > \mathbb{P}(A \mid \overline{B}).$$

(c) If A is independent of B and A is independent of C, must A be independent of $B \cap C$? What about $B \cup C$?

Problem 2.3. Three cards are drawn successively without replacement from a deck of 52 cards. Find the probability that:

(a) all three are aces;

(b) the third card is an ace;

(c) the second card is an ace, given that the first is not.

Problem 2.4. A box contains three coins: two fair coins and one with heads on both sides. We choose a coin uniformly and toss it. Conditional on the chosen coin, successive tosses are independent.

(a) The outcome is heads. What is the probability that the coin is the false one?

(b) We toss the same coin again and get heads again. Find the new probability in two ways: by updating the preceding answer, treating it as a new prior, and directly from the two tosses.

(c) What is the probability of heads on the third toss, given that the first two were heads?

Problem 2.5. *(Test 1, SE 2023)* An urn contains 4 yellow, 5 green and 2 blue pebbles. We select a pebble uniformly. If it is yellow, we roll one standard die; if green, two; if blue, three. The rolls are independent. Given that the sum is 3, what is the probability that the pebble was green?

Problem 2.6. A test gives a positive result for a person with a disease with probability 99% (sensitivity), and a negative result for a healthy person with probability 99% (specificity). The disease affects 0.5% of the population. We choose a person uniformly and test them.

(a) What is the probability that a person with a positive result actually has the disease?

(b) At what disease prevalence is this probability exactly 1/2?

(c) The person is tested again and the result is again positive. If the two tests are conditionally independent given the person's health status, and have the same sensitivity and specificity, what is the probability that the person has the disease? When is this independence assumption questionable?

Additional problems

☺**Problem 2.7.** *(Kahneman, Thinking, Fast and Slow)* An accident occurs at night in a city with a blue and a green taxi company. 85% of the taxis are green and 15% are blue. Assume that a taxi's probability of being involved in an accident does not depend on its company. A witness states that the taxi was blue. A subsequent assessment shows that the witness identifies the colour correctly with probability 80%, both for blue and for green taxis. What is the probability that the taxi really was blue?

Problem 2.8. Three tokens have sides coloured white–white, black–black and white–black, respectively. We choose a token uniformly and place it with a randomly selected side facing up, the two sides being equally likely. Given that the upper side is white, what is the probability that the hidden side is also white?

☺**Problem 2.9.** *(Monty Hall)* A car is behind one of three doors, chosen uniformly. You select a door without knowing where the car is. The host knows its location and always opens a different empty door, then offers you the chance to switch to the remaining closed door.

(a) Should you switch? Does the probability of winning under the "always switch" strategy depend on how the host chooses a door when there are two possibilities?

(b) You chose door 1 and the host opened door 3. When the host can choose between doors 2 and 3, they open door 3 with probability $p \in [0, 1]$. Find the conditional probability that the car is behind door 2. For which p is switching worthwhile?

☺**Problem 2.10.** *(Birthday paradox)* Assume that different people's birthdays are independent and uniformly distributed over 365 days. What is the smallest number of people for which the probability that at least two share a birthday is greater than $1/2$? Use the multiplication rule. Discuss the assumptions of the model.

Problem 2.11. Two fair coins are tossed independently. Let A be "the first coin shows heads", B be "the second coin shows heads", and C be "the coins show the same outcome". Prove that A, B, C are pairwise independent but not mutually independent. Are A and C independent conditional on B?

☺**Problem 2.12.** *(Simpson's paradox)* The following table contains real data on two treatments for kidney stones (Charig et al., *BMJ* 292, 1986; Julious & Mullee, *BMJ* 309, 1994):

	Treatment A	Treatment B
Small stones	93% (81/87)	87% (234/270)
Large stones	73% (192/263)	69% (55/80)
Overall	78% (273/350)	83% (289/350)

Explain why the comparison between the treatments reverses when the groups are combined. Can these data alone establish which treatment is better?

☺**Problem 2.13.** *(Reservoir sampling)* Let A be an array of $n \geq 3$ distinct elements, indexed starting at 1. The following algorithm is executed, with independent random choices in different iterations:

Algorithm 1: Reservoir sampling for one element

```
sample ← A[1];
for i ← 2 to n do
    j ← a random integer, uniformly distributed on {1, ..., i};
    if j = 1 then
        sample ← A[i];
return sample
```

What is the probability that the final value of *sample* is $A[1]$? $A[3]$? $A[n]$? Prove that the algorithm selects an element of the array uniformly.

Problem 2.14. There are $n \geq 1$ urns, each containing m white and k black balls, where $m + k > 0$. We transfer a uniformly chosen ball from the first urn to the second; after adding it, we choose a ball uniformly from the current contents of the second urn and transfer it to the third, and so on. Find the probability that a ball drawn uniformly from the final urn at the end is white.

Challenges and independent research

★ **Problem 2.15.** *(A hidden common cause)* Let E be an event with $\mathbb{P}(E) = q$, where $0 < q < 1$, and let A and B be conditionally independent given the value of $\mathbf{1}_E$. Let

$$\mathbb{P}(A \mid \mathbf{1}_E = e) = a_e \quad \text{and} \quad \mathbb{P}(B \mid \mathbf{1}_E = e) = b_e, \qquad \text{for } e \in \{0, 1\}.$$

(a) Prove that

$$\mathrm{Cov}(\mathbf{1}_A, \mathbf{1}_B) = q(1-q)(a_1 - a_0)(b_1 - b_0).$$

When is this covariance zero? Explain why, for indicators, this is equivalent to independence of A and B.

(b) Assuming $\mathbb{P}(A \cap B) > 0$, find $\mathbb{P}(E \mid A \cap B)$.

(c) Let

$$q = 0.10, \qquad a_0 = 0.02, \quad a_1 = 0.20, \qquad b_0 = 0.03, \quad b_1 = 0.25.$$

Calculate $\mathbb{P}(A)$, $\mathbb{P}(B)$ and $\mathrm{Cov}(\mathbf{1}_A, \mathbf{1}_B)$. Find and compare

$$\mathbb{P}(E), \qquad \mathbb{P}(E \mid A), \qquad \mathbb{P}(E \mid B), \qquad \mathbb{P}(E \mid A \cap B).$$

Check that

$$\mathbb{P}(A \cap B) = 0.00554 \qquad \text{and} \qquad \mathbb{P}(E \mid A \cap B) = \frac{250}{277}.$$

Interpret the results by viewing E as a hidden common cause and A, B as two observed clues to it.

★ **Problem 2.16.** *(An urn of unknown composition)* For a fixed $n \geq 2$, choose K uniformly from $\{0, 1, \ldots, n\}$. Then use an urn containing K red and $n - K$ blue balls. Make two draws from it. Consider separately sampling with and without replacement. Let R_i be the event "the ith ball drawn is red", for $i \in \{1, 2\}$.

(a) Given that the first ball is red, prove that in both models

$$\mathbb{P}(K = k \mid R_1) = \frac{2k}{n(n+1)}, \qquad \text{for } k \in \{0, \ldots, n\}.$$

(b) Find $\mathbb{P}(R_2 \mid R_1)$ in both models and prove that the results are, respectively,

$$\frac{2n+1}{3n} \qquad \text{and} \qquad \frac{2}{3}.$$

(c) Explain probabilistically why the results differ for finite n and why the difference disappears as $n \to \infty$.

Problem 2.17. *(When does a repeated test fail to provide independent evidence?)* Let $D \sim \mathrm{Ber}(0.01)$ indicate the presence of a disease, and let $B \sim \mathrm{Ber}(q)$ indicate a problematic laboratory batch, with D and B independent. All tests on a given person use the same batch. Conditional

on (D, B), the results $T_1, T_2, \ldots$ are independent and identically distributed, with $T_i = 1$ meaning a positive result. Initially let $q = 0.10$, with positive-result probabilities given by

	$B = 0$	$B = 1$
$\mathbb{P}(T_i = 1 \mid D = 1, B)$	0.95	0.80
$\mathbb{P}(T_i = 1 \mid D = 0, B)$	0.05	0.30

(a) Derive an exact formula for

$$p_k = \mathbb{P}(D = 1 \mid T_1 = \cdots = T_k = 1), \qquad \text{for } k \geq 1.$$

(b) Compare it with the naive model that first averages sensitivity and the false-positive probability over B, then assumes that test results are conditionally independent given D. Explain why this assumption can change the answer.

(c) Make a table for $k = 1, \ldots, 5$. Check that

$$p_1 = \frac{17}{152}, \qquad p_2 = \frac{701}{1592} \approx 0.4403,$$

whereas the naive model gives approximately 0.6109 for $k = 2$.

(d) Consider the parameter grid

$$q \in \{0, 0.05, 0.10, 0.20\}, \qquad \mathbb{P}(T_i = 1 \mid D = 0, B = 1) \in \{0.10, 0.20, 0.30, 0.40\},$$

keeping all other parameters fixed. Find the pairs for which the naive estimate of p_2 exceeds the exact value by at least 0.10.

(e) Does the naive model always overestimate p_2 if all four probabilities in the table may vary? Give a proof or a counterexample and explain the result.

(f) Does the naive model always overestimate p_2 if all four probabilities in the table may vary? Find a necessary and sufficient condition for overestimation, give a counterexample, and interpret the condition.

Lean. Let $A, B \subseteq \Omega$, with indicators taking values in $\mathbb{N}$. Use the definition from the first topic and prove in Lean 4 with Mathlib that, for every $x \in \Omega$,

$$\mathbf{1}_{A \cap B}(x) = \mathbf{1}_A(x)\mathbf{1}_B(x).$$

3. Random variables and discrete distributions

Core problems

Problem 3.1. A player stakes 5 leva and rolls two independent standard dice. Two sixes pay 100 leva; exactly one six pays 5 leva; no sixes pay nothing. Let X be the player's net gain. Find the distribution and expectation of X. Is the game fair?

Problem 3.2. A person tosses a fair coin, taking a step forwards for heads and backwards for tails. Toss outcomes are independent. What is the probability that after 10 tosses the person is:

(a) at the starting point;

(b) at distance 2 from it;

(c) five steps ahead of it?

Problem 3.3. Let Z_1 and Z_2 be the outcomes of two independent dice rolls, and let $X = Z_1 + Z_2$.

(a) The dice are standard. Find the distribution, expectation and variance of X.

(b) Let

$$\mathbb{P}(Z_i = 1) = \mathbb{P}(Z_i = 6) = \frac{1}{4} \quad \text{and} \quad \mathbb{P}(Z_i = j) = \frac{1}{8}, \qquad \text{for } i \in \{1, 2\},\ j \in \{2, 3, 4, 5\}.$$

Find the distribution, expectation and variance of X, showing two ways of finding the expectation.

(c) For standard dice, let $Y = |X - 7|$. Find the distribution of Y and $\mathbb{E}Y$. Compare $\mathbb{E}Y$ with $|\mathbb{E}X - 7|$, and $\mathbb{E}Y^2$ with $\operatorname{Var} X$.

Problem 3.4. A box contains 7 light bulbs, 3 of them defective. Four distinct bulbs are chosen uniformly. Find the distribution and expectation of the number of defective bulbs chosen. Also find the expectation using indicators, without summing over the whole distribution. Find the variance using indicators as well. Are these indicators independent?

Problem 3.5. Two independent standard dice are rolled five times, with outcomes in different rolls also independent. What is the probability that exactly 2 rolls have sum 6? Find the expectation and variance of the number of such rolls.

Problem 3.6. *(Memorylessness; Test 1, SE 2023)* Let $X \sim \text{Geom}(p)$, meaning $\mathbb{P}(X = k) = p(1-p)^k$ for $k \in \{0, 1, 2, \dots\}$, where $p \in (0, 1]$.

(a) Prove that for all integers $m, n \geq 0$ with $\mathbb{P}(X \geq n) > 0$,

$$\mathbb{P}(X \geq m + n \mid X \geq n) = \mathbb{P}(X \geq m).$$

(b) Conversely, let Y take values in $\{0, 1, 2, \dots\}$ and satisfy the above equality for every pair of integers $m, n \geq 0$ with $\mathbb{P}(Y \geq n) > 0$. Prove that $Y \sim \text{Geom}(p)$ for some $p \in (0, 1]$.

Problem 3.7. Assume that the number of minor earthquakes in a region over t months has a Poisson distribution with parameter $2t$. What is the probability of fewer than four earthquakes in three months?

Problem 3.8. *(St. Petersburg paradox)* A casino charges an entry fee of A leva. We toss a fair coin until the first heads, with independent outcomes. If heads first occurs on toss n, the prize is 2^n leva. Find the expected prize. For which A would you personally play? What changes if the prize is $\min\{2^n, 2^{30}\}$ leva? How does this result compare with your answer about the values of A for which you would play?

Additional problems

Problem 3.9. An urn contains 5 white and 3 black balls. We draw until a white ball appears. Let X be the number of black balls drawn. Find its distribution, expectation and variance for:

(a) sampling without replacement;

(b) sampling with replacement.

Problem 3.10. 80% of home printers work without additional configuration, with independent outcomes for different printers and weeks. A company sells 10 printers each week. Find the probability that at least 9 work immediately; the probability that this happens in five consecutive weeks; and the probability that the first week in which fewer than 9 work immediately is week k. Calculate the last probability for $k = 3$ and $k = 21$, and find the expected number of weeks until the first such week.

Problem 3.11. *(Test 1, SE 2023)*

(a) Let $X_1, \dots, X_n$ be independent and identically distributed, with $F(m) = \mathbb{P}(X_1 \le m)$, and let $X_{\min} = \min\{X_1, \dots, X_n\}$ and $X_{\max} = \max\{X_1, \dots, X_n\}$. Express $\mathbb{P}(X_{\min} \le m)$ and $\mathbb{P}(X_{\max} \le m)$ in terms of $F(m)$ and n.

(b) Five standard dice are rolled. In each round, we roll all remaining dice simultaneously and remove those showing six. All rolls are independent. Let K be the number of rounds until all dice have been removed. What is the distribution of the number of rolls of one die? Find $\mathbb{P}(K \le n)$ and $\mathbb{P}(K = n)$ for every integer $n \ge 1$.

Problem 3.12. Let X take values in $\{0, 1, 2, \dots\}$ and have finite expectation.

(a) Prove that

$$\mathbb{E}X = \sum_{k=1}^{\infty} \mathbb{P}(X \ge k) = \sum_{n=0}^{\infty} \mathbb{P}(X > n).$$

(b) Check the formula for $X \sim \mathrm{Geom}(p)$.

(c) Find $\mathbb{E}K$ for the number of rounds K in the preceding problem and compare it with the expected number of rolls of one die.

Problem 3.13. Independent Bernoulli trials have success probability p, where $0 < p < 1$. Find the probability that the rth success occurs on exactly trial $k + r$, for integers $r \ge 1$ and $k \ge 0$. Let X_i indicate success on trial i. Express the event using X_i and the partial sums $S_n = \sum_{i=1}^{n} X_i$, with $S_0 = 0$.

Problem 3.14. A shot hits its target with probability 0.001. There are 5000 independent shots, and at least two hits are required. Find the exact probability and the Poisson approximation. How large is the numerical difference?

Problem 3.15. A submarine fires n times; shots are independent and each torpedo hits with probability p. On a hit, one of the m compartments is selected uniformly, independently of other hits. Find the probability that at least two distinct compartments are flooded.

Problem 3.16. *(Banach's matchboxes)* A smoker carries two boxes containing n matches each. Each time, the smoker independently chooses one box with probability $1/2$ and takes a match. On first reaching for an empty box, what is the probability that the other box contains exactly $k \in \{0, \dots, n\}$ matches?

Challenges and independent research

★ **Problem 3.17.** Let π be a uniformly random permutation of $\{1,\dots,n\}$, and let

$$I_n = \#\{(i,j) : 1 \le i < j \le n,\ \pi(i) > \pi(j)\}.$$

Prove that

$$\mathbb{E}I_n = \frac{n(n-1)}{4} \qquad \text{and} \qquad \operatorname{Var}(I_n) = \frac{n(n-1)(2n+5)}{72}.$$

★ **Problem 3.18.** *(Coupon collector)* Let $X_1, X_2, \dots$ be independent and identically distributed, uniform on $\{1,\dots,m\}$, where $m \ge 2$, and let

$$T = \inf\{n \ge 1 : \{X_1,\dots,X_n\} = \{1,\dots,m\}\}.$$

(a) Find $\mathbb{P}(T \le n)$ for $n \ge 1$.

(b) Prove that

$$\mathbb{E}T = mH_m, \qquad \operatorname{Var}(T) = m^2 H_m^{(2)} - mH_m,$$

where $H_m = \sum_{j=1}^{m} \frac{1}{j}$ and $H_m^{(2)} = \sum_{j=1}^{m} \frac{1}{j^2}$.

Where do you use independence of the waiting times, and where is linearity of expectation alone sufficient?

Problem 3.19. *(Generalised coupon collector)* Let $X_1, X_2, \dots$ be independent and identically distributed random variables taking values in $[m] = \{1,\dots,m\}$, where $m \ge 2$, with

$$\mathbb{P}(X_n = i) = p_i > 0, \qquad \text{for } i \in [m], \qquad \text{and } \sum_{i=1}^{m} p_i = 1.$$

Let T be the number of draws until all m values have appeared.

(a) Prove that

$$\mathbb{E}T = \sum_{\emptyset \ne A \subseteq [m]} \frac{(-1)^{|A|+1}}{\sum_{i \in A} p_i} \qquad \text{and} \qquad \mathbb{E}T = \int_0^\infty \left(1 - \prod_{i=1}^{m} (1 - e^{-p_i t})\right) dt.$$

Explain why the first formula is unsuitable for direct numerical calculation when $m = 30$. Suggest a numerical method for the integral and a way to check its accuracy.

(b) Compare values from the integral formula with simulations for $m = 10, 30, 100$ in the following models:

(i) uniform probabilities: $p_i = 1/m$ for $i \in [m]$;

(ii) probabilities

$$p_i = \frac{i^{-\alpha}}{\sum_{j=1}^{m} j^{-\alpha}}, \qquad \text{for } i \in [m],$$

considering both $\alpha = 1$ and $\alpha = 2$;

(iii) one rare coupon:

$$p_1 = m^{-2} \quad \text{and} \quad p_i = \frac{1 - m^{-2}}{m-1}, \qquad \text{for } i \in \{2,\dots,m\}.$$

State the number of simulations and discuss the accuracy of the estimated mean. Explain which mechanism determines the scale of $\mathbb{E}T$ in each model.

You may use AI for the proof and numerical method, provided you check the results and explain how you checked them.

Lean. Let S be a finite nonempty set, $X, Y : S \to \mathbb{R}$, and $a, b \in \mathbb{R}$. On the finite uniform probability space on S, define

$$\mathbb{E}X = \frac{1}{|S|} \sum_{s \in S} X(s).$$

Formalise in Lean 4 with Mathlib and prove linearity:

$$\mathbb{E}(aX + bY) = a\,\mathbb{E}X + b\,\mathbb{E}Y.$$

First prove, by induction on S and without using `Finset.sum_add_distrib`, that

$$\sum_{s \in S} \big(X(s) + Y(s)\big) = \sum_{s \in S} X(s) + \sum_{s \in S} Y(s);$$

`Finset.mul_sum` and `Finset.sum_div` are useful for the rest. Why is nonemptiness of S needed to define $\mathbb{E}$, but not for the identity for sums?

4. Dependence, sums and generating functions

Part I. Joint distributions and conditional expectation

Problem 4.1. We independently roll two red and one blue standard dice. Let X be the number of sixes on the red dice, and Y the number of twos on all three dice. Find the joint distribution and $\mathbb{P}(X > 0 \mid Y = 1)$. Are X and Y independent? Can you answer the last question without calculating the entire joint distribution?

Problem 4.2. A fair coin is tossed independently four times. Let X be the number of heads in the first three tosses, and Y the number in the last two.

(a) Find the joint and conditional distributions, $\mathbb{P}(X = Y)$ and $\mathbb{P}(X > 1 \mid Y = 1)$.

(b) Find $\mathbb{E}(X \mid Y)$ and $\mathbb{E}(Y \mid X)$ and their distributions. Verify that $\mathbb{E}[\mathbb{E}(X \mid Y)] = \mathbb{E}X$.

(c) Find $\operatorname{Cov}(X, Y)$ in two ways: from the joint distribution, and by writing X and Y as sums of indicators.

Problem 4.3. Let X be uniformly distributed on $\{-1, 0, 1\}$ and $Y = X^2$.

(a) Show that $\operatorname{Cov}(X, Y) = 0$, but X and Y are dependent.

(b) Now let X and Y take values in $\{0, 1\}$. Prove that $\operatorname{Cov}(X, Y) = 0$ is equivalent to independence. Why does this not contradict the preceding example?

(c) For random variables X, Y with finite positive variances, prove that $|\operatorname{Cor}(X, Y)| \leq 1$. When does equality hold?

Problem 4.4. One box contains 4 white and 6 black balls, and another contains 7 white and 3 black balls. From each box we draw one ball uniformly, independently of the other draw. We add one white ball to the two drawn balls and place all three in a third box.

(a) What is the probability that a uniformly chosen ball from the third box is white?

(b) Draw from the third box with replacement until the first white ball. For a fixed box composition, draws are independent and uniform. Let X be the number of draws. Find $\mathbb{E}X$. Does X have a geometric distribution?

(c) Find $\mathbb{P}(X > k)$ for every integer $k \geq 0$, and prove that $\mathbb{P}(X > k + 1 \mid X > k)$ increases with k. What do we learn about the box's composition after a sequence of black balls?

Additional problems

Problem 4.5. Choose three numbers uniformly without replacement from $\{1, 2, 3, 4, 5\}$. Let X be the middle number in order of size, and Y the smallest.

(a) Find the joint and marginal distributions. Are X, Y independent?

(b) Find $\operatorname{Cov}(X, Y)$ and $\operatorname{Cor}(X, Y)$.

(c) Find the distribution, expectation and variance of $X - 2Y$.

Problem 4.6. Let $(X_1, \ldots, X_n)$ be a uniformly random permutation of $\{1, \ldots, n\}$, where $n \geq 2$, and let $S = X_1 + \cdots + X_n$.

(a) Find $\mathbb{E}S$, $\operatorname{Var} S$, $\mathbb{E}X_i$ and $\operatorname{Var} X_i$.

(b) Prove the variance-of-a-sum formula and use it to find $\operatorname{Cov}(X_i, X_j)$ for $i \neq j$. Explain the sign of the covariance.

(c) For $1 \le k \le n$, find $\mathrm{Var}(X_1 + \cdots + X_k)$. Compare with the variance of k independent draws with replacement.

Problem 4.7. On each face of a blank die, independently write a uniformly chosen number from 1 to 6. Then roll the resulting die $n \ge 2$ times. The faces rolled are independent, uniformly selected and independent of the labels. Let T_k be the number seen on roll k.

(a) What is the distribution of T_1? Find $\mathrm{Cor}(T_1, T_2)$. Are T_1, T_2 independent?

(b) If $X = T_1 + \cdots + T_{n-1}$ and $Y = T_2 + \cdots + T_n$, find $\mathrm{Cor}(X, Y)$.

(c) How do the two correlations change if the labels are independent and uniform on $\{199, \ldots, 999\}$? What if the labels are independent with an arbitrary common distribution having finite positive variance?

Problem 4.8. *(Pólya's urn)* An urn contains one red and one black ball. At each step, choose a ball uniformly, replace it and add another ball of the same colour. Let X_n indicate a red ball on draw n, and let $W_n = X_1 + \cdots + X_n$. Find $\mathbb{P}(X_n = 1)$, $\mathrm{Cov}(X_i, X_j)$ for $i \ne j$, and the distribution of W_4. What distribution do you conjecture for W_n?

Challenges

★ **Problem 4.9.** *(Pairwise independence)* Let q be prime, and let A, B be independent and uniform on $\{0, \ldots, q-1\}$. All subsequent calculations are modulo q. For $t \in \{0, \ldots, q-1\}$, define

$$X_t = A + tB \pmod{q}, \qquad X_\infty = B.$$

(a) Prove that $X_0, \ldots, X_{q-1}, X_\infty$ are uniform and pairwise independent, but not mutually independent. What do the values of two distinct coordinates determine?

(b) Let N be the number of zero coordinates in this family. Find its distribution, expectation and variance.

(c) Calculate the variance using indicators as well. Is pairwise independence enough for variances to add?

★ **Problem 4.10.** For random variables X, Y with finite second moments and a random variable Z, prove that

$$\mathrm{Cov}(X, Y) = \mathbb{E}[\mathrm{Cov}(X, Y \mid Z)] + \mathrm{Cov}(\mathbb{E}[X \mid Z], \mathbb{E}[Y \mid Z]) .$$

What formula results when $Y = X$? Now let $Z \sim \mathrm{Ber}(1/2)$, let U be uniform on $\{-1, 1\}$, let Z, U be independent, and define

$$X = 3Z + U, \qquad Y = 3Z - U.$$

(a) Find the conditional covariance and correlation for fixed Z.

(b) Find $\mathrm{Cov}(X, Y)$ and $\mathrm{Cor}(X, Y)$.

(c) How do you explain the difference in signs? Relate your answer to the two terms in the formula.

Part II. Generating functions and random sums

Problem 4.11. *(Test 1, CS 2024)* For a nonnegative integer-valued random variable X, write $g_X(s) = \mathbb{E}[s^X]$, $0 \le s \le 1$.

(a) Let X be uniform on $\{1, \ldots, 6\}$. Find g_X and express $\mathbb{E}X$ and $\mathrm{Var}\, X$ in terms of its derivatives.

(b) If X, Y are independent nonnegative integer-valued random variables, prove that $g_{X+Y} = g_X g_Y$.

(c)* Is the converse true?

Problem 4.12. Let X, Y be independent. Use generating functions to find the distribution of $X+Y$ when $X \sim \mathrm{Bin}(n,p)$ and $Y \sim \mathrm{Bin}(m,p)$, and then when $X \sim \mathrm{Poi}(\lambda)$ and $Y \sim \mathrm{Poi}(\mu)$. Similarly find the distribution of $G_1 + \cdots + G_r$, where $G_1, \ldots, G_r$ are independent and $G_i - 1 \sim \mathrm{Geom}(p)$, so each G_i counts the trials up to the first success. Why does equality of generating functions determine the distribution uniquely? Can the sum of independent $\mathrm{Bin}(n,p)$ and $\mathrm{Bin}(m,q)$ variables be binomial when $n, m \geq 1$, $0 < p, q < 1$ and $p \neq q$?

Problem 4.13. The number of visitors to a stadium is $N \sim \mathrm{Poi}(\lambda)$. Each visitor selects one of ten entrances uniformly, independently of everyone else and of N.

(a) Find the distribution, expectation and variance of the number using the first entrance.

(b) More generally, let $N, X_1, X_2, \ldots$ be independent nonnegative integer-valued random variables, with (X_i) identically distributed. For $S = X_1 + \cdots + X_N$, setting $S = 0$ when $N = 0$, prove that

$$g_S(s) = g_N(g_{X_1}(s)).$$

If N, X_1 have finite second moments, find $\mathbb{E}S$ and $\operatorname{Var} S$.

(c) Apply the result to the stadium problem. Also obtain the distribution in (a) using the law of total probability.

Problem 4.14. A and B play independent games. In each game, A wins with probability $2/3$ and B with probability $1/3$. Play stops when someone wins two consecutive games. Let T be the number of games played. Find $\mathbb{E}T$ and the probability that A wins the contest. What information about the past suffices to describe future play? Also find the generating function of T.

Additional problems

Problem 4.15. Two teams each take one penalty in each round. All attempts are independent, with scoring probabilities 0.75 for the first team and 0.80 for the second. All five regular rounds are played. If tied afterwards, play continues until, in one round, one team scores and the other does not.

(a) What is the probability that in each of the first five rounds both teams have the same outcome for their penalty?

(b) What is the probability that play continues after the fifth round? After the tenth?

(c) Find the expected total number of penalties taken.

Problem 4.16. Lottery tickets are numbered from 000000 to 999999. Find the probability that a uniformly chosen ticket:

(a) has digit sum 21;

(b) has equal sums of its first and last three digits;

(c) has a sum of its first three digits that exceeds that of its last three by 2.

It suffices to express the answers as coefficients of a polynomial.

Problem 4.17. We telephone $N \sim \mathrm{Poi}(\lambda)$ people with an offer. Person i would accept on the K_ith call, where $K_1, K_2, \ldots$ are independent, identically distributed and independent of N, with $\mathbb{P}(K_i = n) = f(n)$ for $n \geq 1$ and $\sum_{n \geq 1} f(n) = 1$. Let X_n be the number of people who would accept on call n.

(a) Prove that $(X_n)_{n \geq 1}$ are independent and $X_n \sim \mathrm{Poi}(\lambda f(n))$.

(b) We stop at the first round with no new acceptance. If $T = \inf\{n \geq 1 : X_n = 0\}$ and $S = X_1 + \cdots + X_T$, prove that

$$\mathbb{E}S = \lambda \mathbb{E}[F(T)], \qquad F(k) = \sum_{j=1}^{k} f(j).$$

Why can we not directly apply the expectation formula for a random sum whose number of summands is independent of the summands?

Problem 4.18. Two standard dice are rolled independently. Can we label their faces with other positive integers so that independent rolls have the same sum distribution? Repeated labels are allowed. Find a construction different from the two standard dice.

Challenges

Problem 4.19. We toss a fair coin independently, recording outcomes as H, T. For a word $w = w_1 w_2 w_3$, let τ_w be the toss number when it first appears as three consecutive outcomes. Consider $a = HHT$ and $b = THH$.

(a) Choose states retaining enough information about the appearance of the two words, and draw the possible transitions.

(b) Find $\mathbb{P}(\tau_a < \tau_b)$ and $\mathbb{E}(\tau_a \wedge \tau_b)$.

(c) Prove that $\tau_a \wedge \tau_b < \infty$ almost surely, without using the calculated expectation.

Problem 4.20. *(Branching process)* A population starts with one individual. Each individual has 0 children with probability $1 - p$ and 2 children with probability p, where $0 < p < 1$, independently of all other individuals in all generations. Let Z_n be the number of individuals in generation n, with $Z_0 = 1$, and let

$$f(s) = 1 - p + ps^2, \qquad T = \inf\{n \geq 0 : Z_n = 0\}.$$

(a) Prove that $g_{Z_n} = f^{\circ n}$.

(b) Show that $q_n = \mathbb{P}(T \leq n) = f^{\circ n}(0)$ and that (q_n) is increasing.

(c) Prove that $q = \mathbb{P}(T < \infty)$ is the smallest solution of $f(q) = q$ in $[0, 1]$, and find it as a function of p.

(d) Find $\mathbb{E}Z_n$. When $p = 1/2$, it is constant, yet extinction occurs almost surely. How do you explain this?

Independent research

Problem 4.21. *(Penney's game)* Two players each choose a different word from $\{H, T\}^3$. We then toss a fair coin independently until one of the words appears. The player whose word appears first wins.

(a) Generalise the model in Problem 4.19 and write a program that calculates exactly, for each of the 56 ordered pairs (a, b),

$$p(a, b) = \mathbb{P}(\tau_a < \tau_b), \qquad e(a, b) = \mathbb{E}(\tau_a \wedge \tau_b).$$

Use rational arithmetic and check at least one pair by hand.

(b) For each first word a, find the best responses b for the second player. Does the player choosing second have an advantage? Formulate a general rule for the best response and prove it.

(c) Draw a directed graph with edge $a \to b$ when b is a best response to a. Find a cycle and explain what it shows about comparisons between words.

Lean. Formalise in Lean 4 with Mathlib and prove that, for finite sets S, T and functions $a : S \to \mathbb{R}$ and $b : T \to \mathbb{R}$,

$$\sum_{s \in S} \sum_{t \in T} a(s) b(t) = \left(\sum_{s \in S} a(s) \right) \left(\sum_{t \in T} b(t) \right).$$

Explain how this identity is used in proving $g_{X+Y} = g_X g_Y$ for independent nonnegative integer-valued random variables X, Y with finite ranges.